

```

In [4]: # Part 1: Tutorial

# import the NumPy Library
import numpy

# transition probability matrix
A = numpy.array([[0, 0.25, 0.25, 0.25, 0.25],
                 [0.5, 0, 0, 0.5, 0],
                 [0.33, 0, 0, 0.33, 0.33],
                 [1, 0, 0, 0, 0],
                 [0, 0, 0, 1, 0]])

# initialize the PageRank vector
v = numpy.array([[0.2], [0.2], [0.2], [0.2], [0.2]])

# the damping factor
d = 0.85

# the size of the "Internet"
N = 5

# compute the update matrix
U = d * A.T + (1 - d) / N

# compute the new PageRank vector
v = numpy.dot(U, v)

# print the new PageRank vector
print(v)

# initialize the PageRank vector
v = numpy.array([[0.2], [0.2], [0.2], [0.2], [0.2]])

# print the initial vector
print('PageRank vector', 0, 'is', v.T)

# compute the PageRank vector for 15 iterations
for i in range(15):
    # compute the next PageRank vector
    v = numpy.dot(U, v)

    # round the PageRank vector to 3 places
    v = numpy.round(v, 3)

    # print the PageRank vector
    print('PageRank vector', i + 1, 'is', v.T)

```

```

[[0.3411]
 [0.0725]
 [0.0725]
 [0.3836]
 [0.1286]]
PageRank vector 0 is [[0.2 0.2 0.2 0.2 0.2]]
PageRank vector 1 is [[0.341 0.073 0.073 0.384 0.129]]
PageRank vector 2 is [[0.408 0.102 0.102 0.264 0.123]]
PageRank vector 3 is [[0.326 0.117 0.117 0.293 0.145]]
PageRank vector 4 is [[0.362 0.099 0.099 0.305 0.132]]
PageRank vector 5 is [[0.359 0.107 0.107 0.289 0.135]]
PageRank vector 6 is [[0.351 0.106 0.106 0.296 0.136]]
PageRank vector 7 is [[0.356 0.104 0.104 0.295 0.134]]
PageRank vector 8 is [[0.354 0.105 0.105 0.293 0.135]]
PageRank vector 9 is [[0.353 0.105 0.105 0.294 0.134]]
PageRank vector 10 is [[0.354 0.105 0.105 0.293 0.134]]
PageRank vector 11 is [[0.353 0.105 0.105 0.293 0.134]]
PageRank vector 12 is [[0.353 0.105 0.105 0.293 0.134]]
PageRank vector 13 is [[0.353 0.105 0.105 0.293 0.134]]
PageRank vector 14 is [[0.353 0.105 0.105 0.293 0.134]]
PageRank vector 15 is [[0.353 0.105 0.105 0.293 0.134]]

```

```

In [5]: # Page | Links to | Out-degree (C)
# -----|-----|-----
# 0 | 2, 3, 4 | 3
# 1 | 4 | 1
# 2 | 1, 3 | 2
# 3 | 1 | 1
# 4 | 0, 1, 2 | 3

```

```

In [8]: import numpy

# page 0 --> 2, 3, 4 (C = 3, so 1/3 to each)
# page 1 --> 4 (C = 1, so 1 to page 4)
# page 2 --> 1, 3 (C = 2, so 1/2 to each)
# page 3 --> 1 (C = 1, so 1 to page 1)
# page 4 --> 0, 1, 2 (C = 3, so 1/3 to each)

A = numpy.array([[0,      0,      1/3,    1/3,    1/3],
                 [0,      0,      0,      0,      1],
                 [0,      1/2,    0,      1/2,    0],
                 [0,      1,      0,      0,      0],
                 [1/3,    1/3,    1/3,    0,      0]])

# PageRank vec
v = numpy.array([[0.2], [0.2], [0.2], [0.2], [0.2]])

# damping factor
d=0.6

# size of the "Internet"
N = 5

# compute the update matrix
U= d * A.T + (1 - d) / N

```

```
# print initial PageRank vector
print('PageRank vector', 0, 'is', v.T)

# compute the PageRank vector for 15 iterations

for i in range(15):

    v= numpy.dot(U, v)
    v= numpy.round(v, 3)

    print('pagerank vec', i+1, '---->', v.T)
```

```
PageRank vector 0 is [[0.2 0.2 0.2 0.2 0.2]]
pagerank vec 1 ----> [[0.12 0.3 0.16 0.18 0.24]]
pagerank vec 2 ----> [[0.128 0.284 0.152 0.152 0.284]]
pagerank vec 3 ----> [[0.137 0.274 0.162 0.151 0.276]]
pagerank vec 4 ----> [[0.135 0.274 0.163 0.156 0.272]]
pagerank vec 5 ----> [[0.134 0.277 0.161 0.156 0.271]]
pagerank vec 6 ----> [[0.134 0.276 0.161 0.155 0.273]]
pagerank vec 7 ----> [[0.135 0.276 0.161 0.155 0.272]]
pagerank vec 8 ----> [[0.134 0.276 0.161 0.155 0.273]]
pagerank vec 9 ----> [[0.135 0.276 0.161 0.155 0.272]]
pagerank vec 10 ----> [[0.134 0.276 0.161 0.155 0.273]]
pagerank vec 11 ----> [[0.135 0.276 0.161 0.155 0.272]]
pagerank vec 12 ----> [[0.134 0.276 0.161 0.155 0.273]]
pagerank vec 13 ----> [[0.135 0.276 0.161 0.155 0.272]]
pagerank vec 14 ----> [[0.134 0.276 0.161 0.155 0.273]]
pagerank vec 15 ----> [[0.135 0.276 0.161 0.155 0.272]]
```